Distributed Systems

Monsoon 2024

Lecture 17

International Institute of Information Technology

Hyderabad, India

Reading material:
Karger et al., STOC 1997

How to Locate Data

- A distributed system offers a good abstraction to store and access data.
- Such a system can have commodity machines store data and serve data to other users.
- One way to enable such a system is to provide a simple interface to query the system for data.
- Data itself is stored as key-value pairs.

Distributed Data: Evolution

- The first generation systems that store and serve data usually had some degree of centralization.
- Servers store an index file.
- Example of such a system is Napster.

- The second generation systems moved out of centralization.
- Peers work collaboratively to locate keys and disseminate information.
- Two kinds of systems: structured and unstructured.
- Such systems also known as overlay networks.

Distributed Data: Evolution

- Third generation:
  - In addition to being an overlay, focus is also on other features such as security/anonymity/ and the like.
  - The user and the data server should not be revealing their identity.

Summary of Evolution of Distributed Data Storage Systems

Nature	Main Features	Example Systems
First Generation	Limited decentralized data storage Index file at servers	Napster
Second Generation	Fully decentralized, more peer-to-peer Peers collaborate to store/ locate data Structured Vs Unstructured	Chord Tapestry
Third Generation	Additional features such as security/ anonymity/	

A First Generation System: Napster

- A search engine dedicated to MP3 files.
- Users could share MP3 files with each other.
- Each user on connecting to Napster shares the list of the MP3 files they own with the Napster broker.
- A user can search for MP3 files through the broker which maintains an index.
- The actual file transfer happens between the client that has the file and the client that needs the file.
    - Broker not involved in the actual transfer.

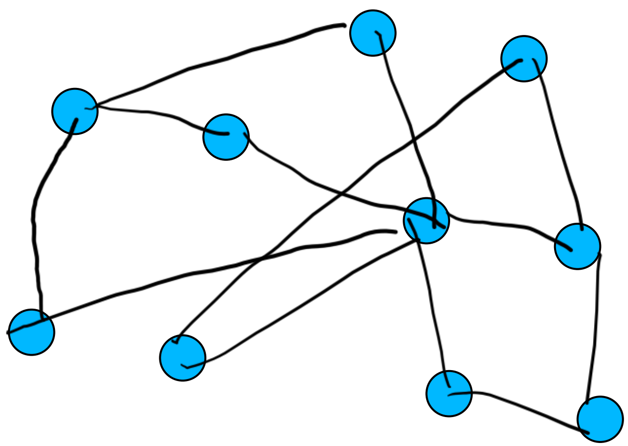
Napster

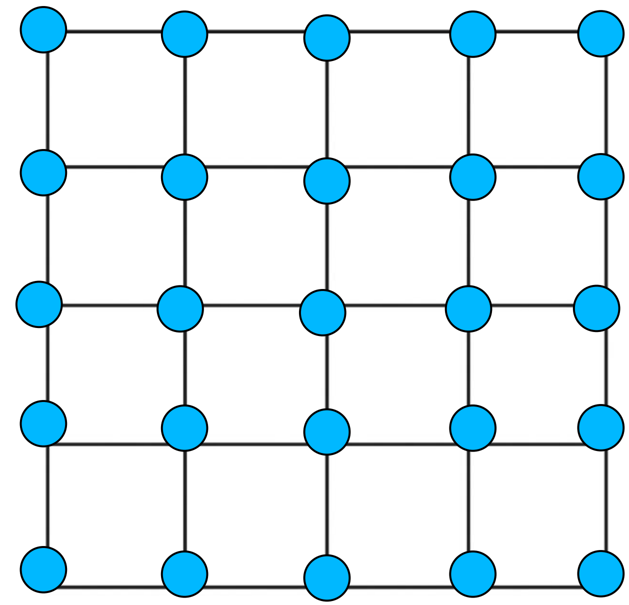
- Multiple broker servers can exist at the same time.
- Several optimizations exist
    - Clients can connect to the least loaded broker.
    - Download also possible from machines that are inside a private network.
- However, the legality of the entire sharing service was heavily up for question.
- Plus, there is no anonymity for any entity.
- Napster was shut down in 2001 due to legal troubles over copyright issues.

Second Generation Systems

- No centralized server.
- Peers have to share information with each other to locate and download shared files.
- Two strategies in general
  - Unstructured networks
  - Structured networks

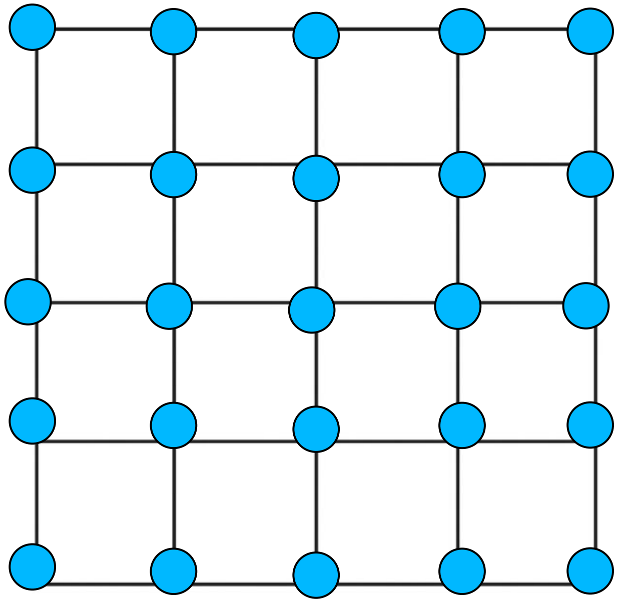
Distributed Data: Evolution

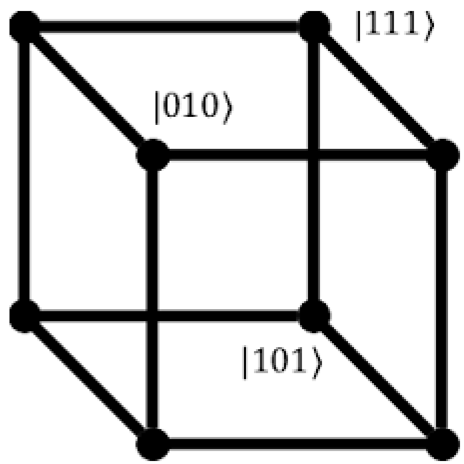


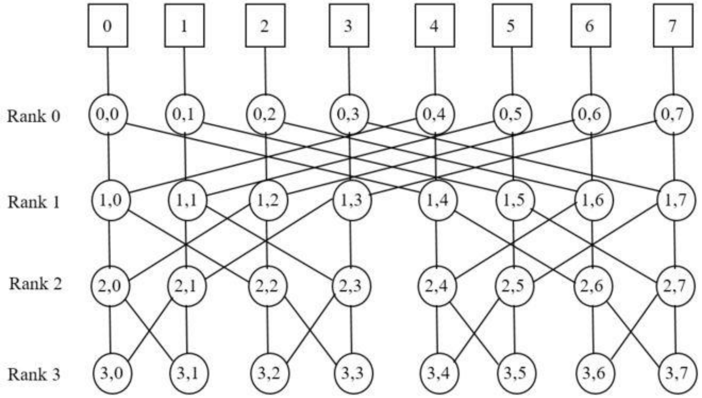


- Unstructured Systems:
  - Arbitrary set of connections between nodes.
  - Information dissemination via gossiping style protocols.
  - Rumor spreading/ epidemic propagation
- Structured Systems:
  - Connections are well-defined. Logically at least.
  - Known as overlay networks.

Structured Vs. Unstructured

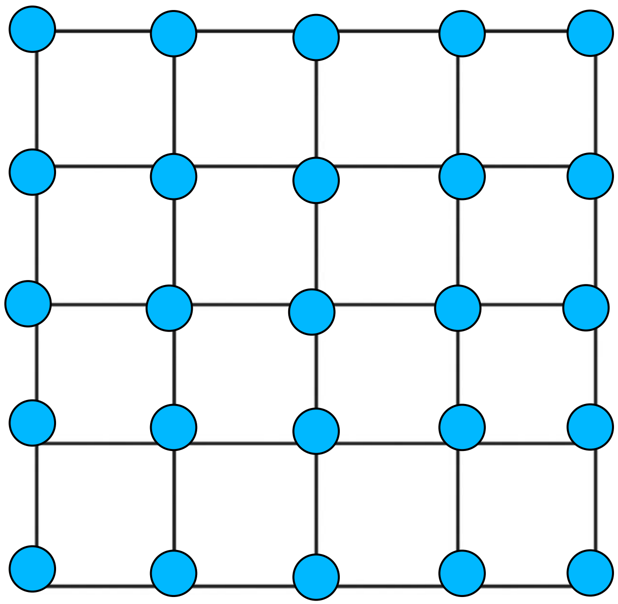


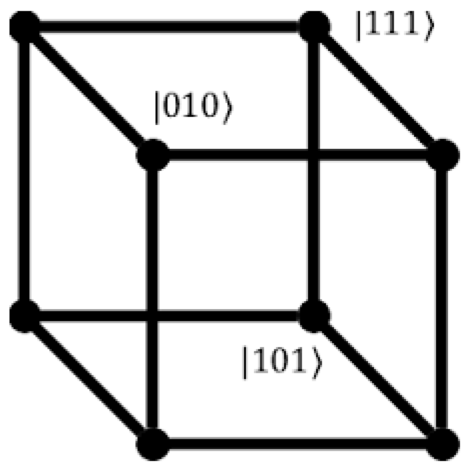


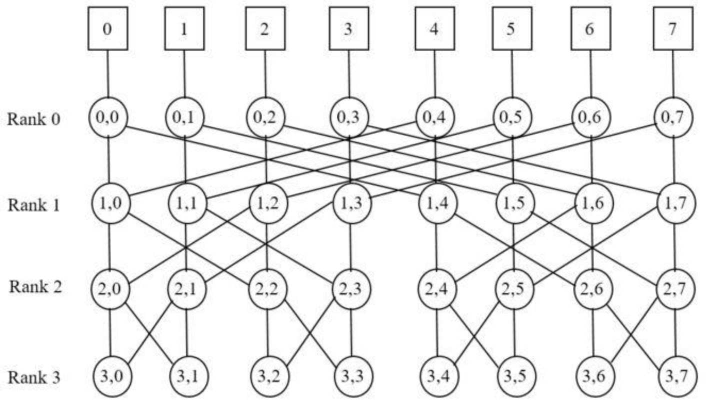


- Structured P2P overlays: These follow a regular topology.
- Example topologies include the mesh, hypercube, butterfly, and such graphs.

Structured Vs. Unstructured

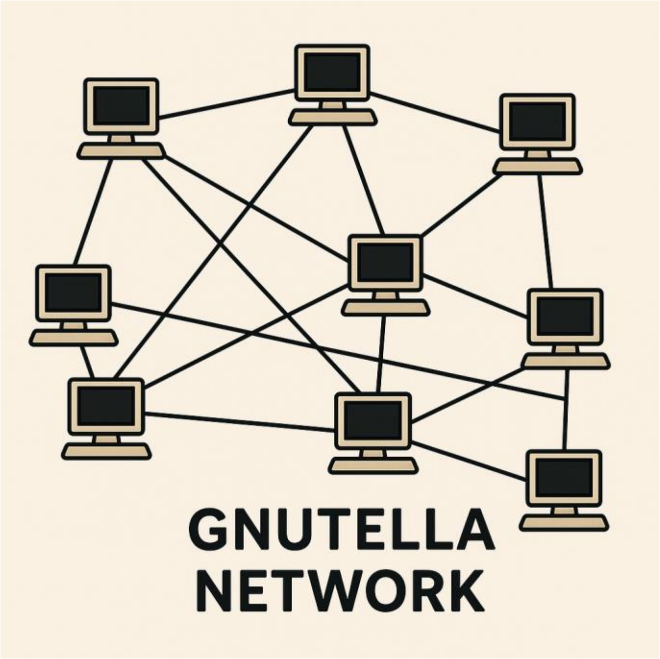






- The topology also guides the procedure for object storage and object search.
  - Object storage and search strategies are intricately linked to the overlay structure as well as to the data organization mechanisms.

Structured Vs. Unstructured



- Unstructured P2P overlays: These do not follow any regular network topology.
  - Loose guidelines for object search and storage
  - Search mechanisms are ad-hoc, variants of flooding and random walk

Characteristics of P2P Networks

- P2P network is an application-level organization of nodes to form a network.
- Such a network allows nodes to flexibly share resources.
- The name P2P comes from the idea that all nodes are equal; communication directly between peers (no client-server)
- The network allows location of arbitrary objects; no DNS-type service is required.
- The network can be seen as a large combined storage, CPU power, other resources, without scalability costs.
- Dynamic insertion and deletion of nodes, as well as of resources, at low cost.

How to Place Data

- Data identified by indexing, which allows physical data independence from applications.
- Three types of indexing possible
  - Centralized indexing, e.g. like in DNS servers
  - Distributed indexing.
    - Indexes to data scattered across peers.
    - Access data through mechanisms such as Distributed Hash Tables (DHT).
    - These differ in hash mapping, search algorithms, diameter for lookup, fault tolerance, churn resilience.
  - Local indexing.
    - Each peer indexes only the local objects.
    - Remote objects need to be searched for. Used commonly in unstructured overlays (E.g., Gnutella) along with flooding search or random walk search.

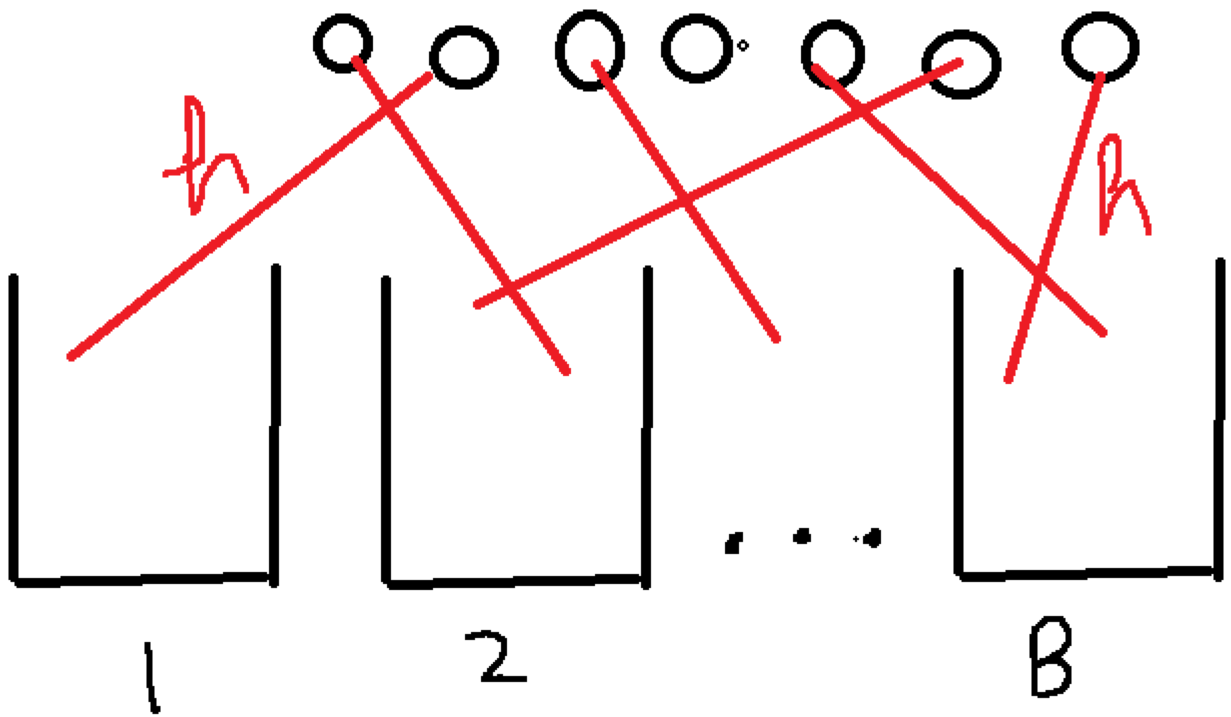
How to Place Data

- Another classification is via semantic vs. semantic-free nature of indexing.
- Semantic indexing is human readable, e.g., filename, keyword, database key.
    - Supports keyword searches, range searches, approximate searches.
- Semantic-free indexing is not human readable.
    - Corresponds to index obtained by use of good hash functions to help in the search process.

How to Place Data – Hashing

- Consider a single server that has many objects and clients that access these objects.
- One often uses caching, and possibly a set of caches, to improve the performance of the server.
- We hope that the objects are distributed uniformly across the caches, so that each cache is responsible for a nearly equal share.
- In addition, clients need to know which cache to query for a specific object.

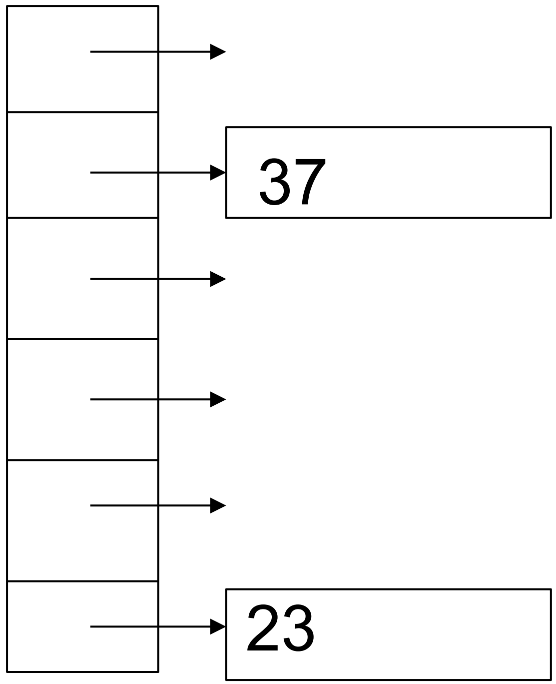
How to Place Data



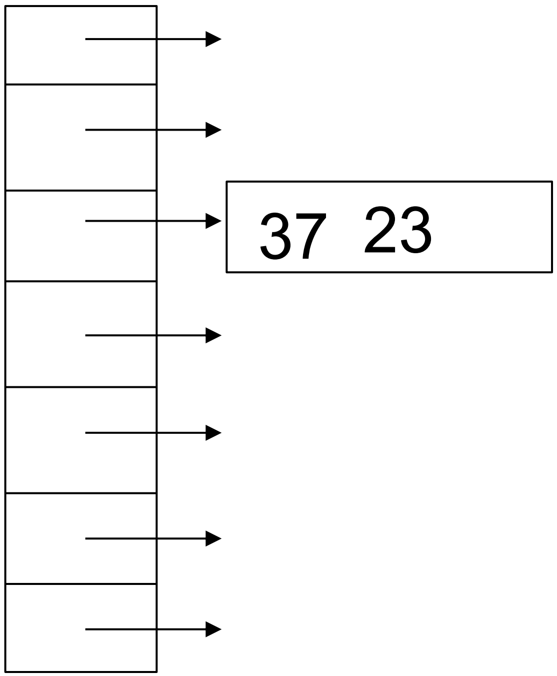
- Hashing provides the following solution.
- The server can use a hash function that evenly distributes the objects across the caches.
- Clients can use the same hash function to discover which cache stores a object.

How to Place Data

p = 6
a = 1
b = 0

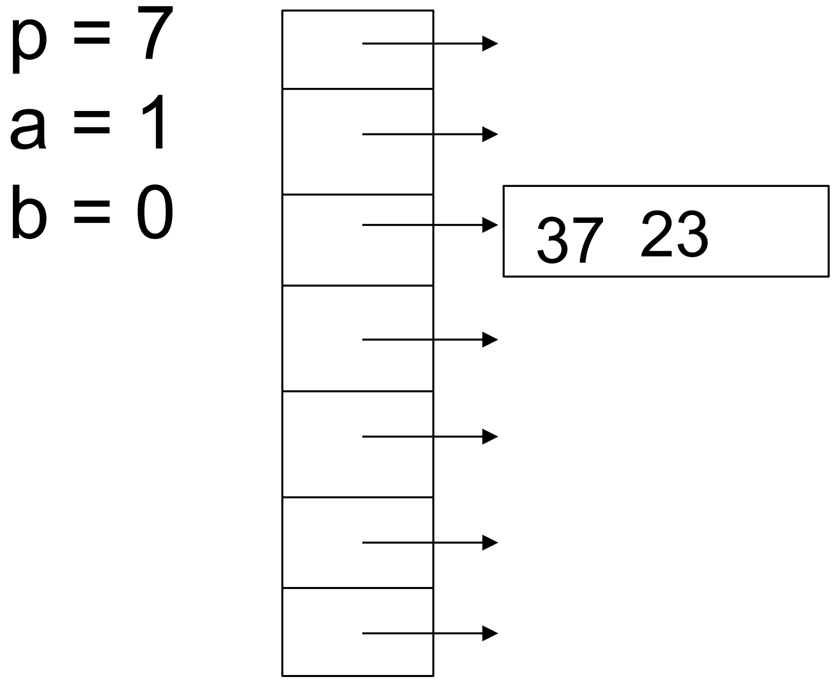


p = 7
a = 1
b = 0



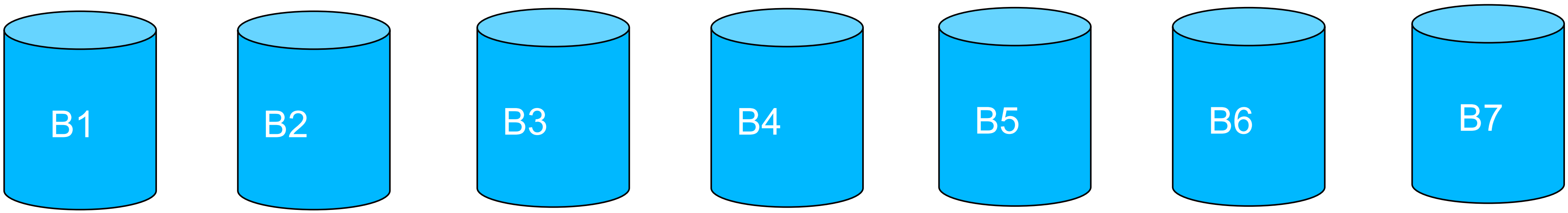
- The solution however has certain drawbacks.
- Consider what happens when the machines that work as caches change.
- Suppose the hash function uses a function modulo p, e.g. f(x) = ax + b (mod p), where p is the number of machines that cache objects.

How to Place Data



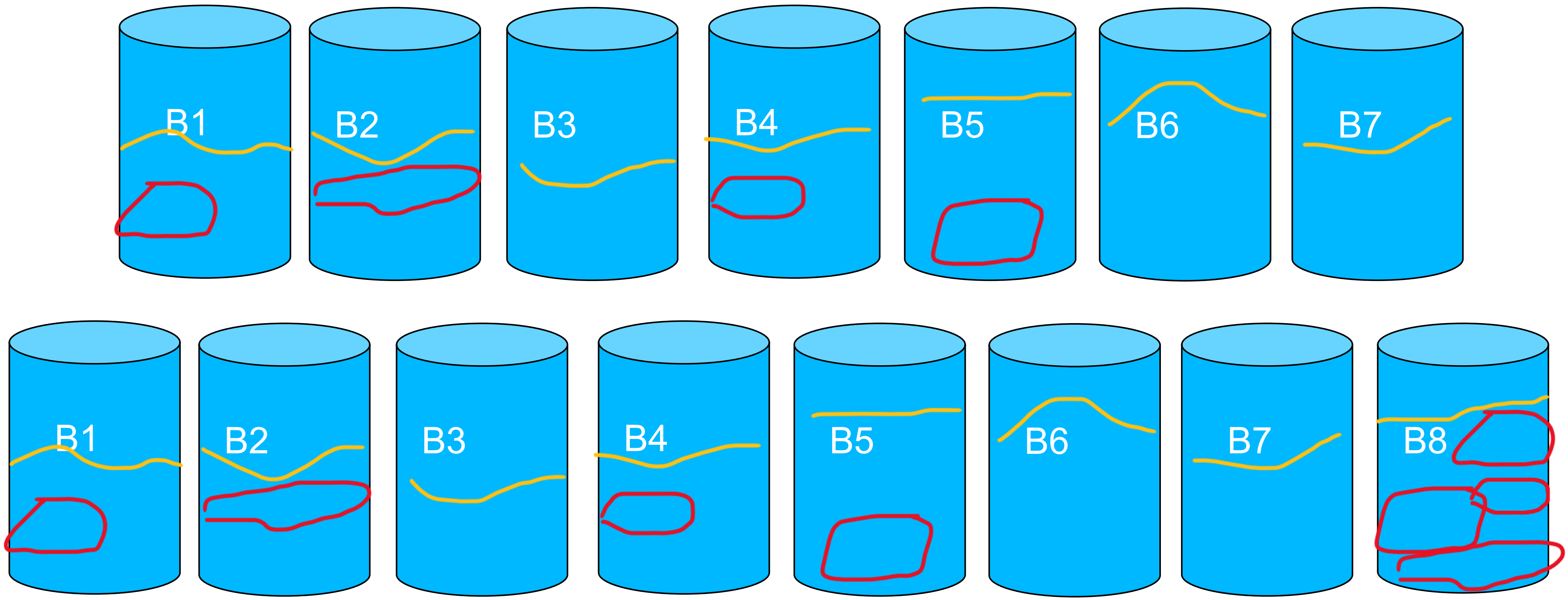
- The solution however has certain drawbacks.
- Consider what happens when the machines that work as caches change.
- When the number of machines change, the range of the hash function (p in the example) has to change.
- In that case, almost every item would be hashed to a new location.

Consistent Hashing – A Solution



- Define a *view* to be the set of buckets of which a particular client is aware.
- View V1 = {C2, C6, C7}
- View V2 = {C1, C3, C5}

Consistent Hashing – A Solution



- Consider hash functions that have the following properties:
- Smoothness: When a machine is added to/removed from the set of buckets, the expected fraction of objects that must be moved to a new bucket is the minimum needed to maintain a balanced load across the buckets.

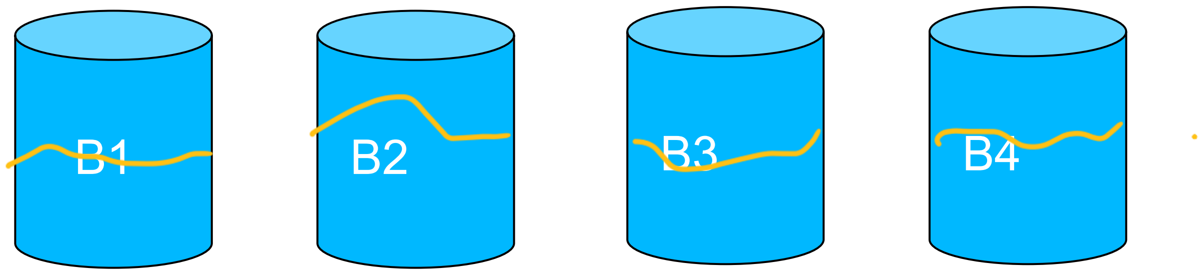
Consistent Hashing – A Solution

View V1 = {B2, B6, B7}
View V2 = {B1, B3, B5}
View V3 = {B2, B6, B7}
View V4 = {B1, B2, B4}

Object id, x = 12
H(x, V1) = 1
H(x, V2) = 4
H(x, V3) = 1
H(x, V4) = 2

- Consider hash functions that have the following properties:
- Spread: Over all the client views, the total number of different buckets to which an object is assigned is small.

Consistent Hashing – A Solution



- Consider hash functions that have the following properties:
- Load: The number of distinct objects assigned to a particular bucket is small.
- Such hash functions are called consistent hash functions.

Consistent Hashing – A Solution

- The “smoothness” property implies that small changes in the set of buckets require only small changes in the location of the objects.
- The “spread” property implies that even in the presence of inconsistent views of the world, references for a given object are directed only to a small number of buckets
- The “load” property implies that each bucket is assigned nearly the same number of objects.

- Do such hashing schemes exist?

A Construction

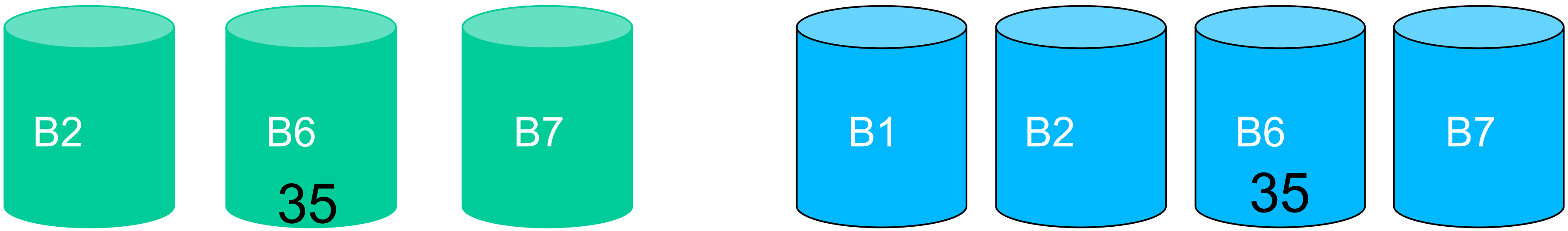
- Let I be the set of items and B be the set of buckets.
- A view V is any subset of the buckets B.
- A ranged hash function is a function of the form $f : 2^B \times \mathrm{I} \rightarrow B$.
  - Such a function specifies an assignment of items to buckets for every possible view.
  - $f(V, i)$ is the bucket to which item i is assigned in view V. Written as $f_V(i)$ from now on.
- A family of such ranged hash functions is called as a ranged hash family.
- We call a random range hash function as a function drawn uniformly at random from a ranged hash family of functions.

A Construction

- We consider ranged hash families that have the following properties.

- Balance: A ranged hash family is balanced if, given a particular view V, a set of items, and a randomly chosen function selected from the hash family, with high probability the fraction of items mapped to each bucket is O(1/|V|).
- The balance property requires that the functions in the hash family distribute items among buckets in a balanced fashion.

A Construction



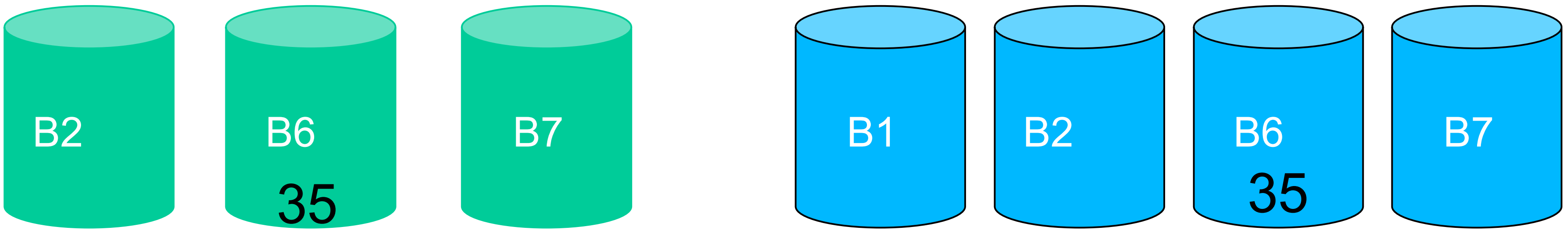
View **V1 = {B2, B6, B7}**

View **V2 = {B1, B2, B6, B7}**

- We consider ranged hash families that have the following properties.
- Monotonicity: A ranged hash function f is monotone if for all views $V_1 \subseteq V_2 \subseteq B$, $f_{V_2}(i) \in V_1$ implies that

  $f_{V_1}(i) = f_{V_2}(i)$.
- If every function from a ranged hash family is monotone, then the family is called a monotone family.

A Construction



View **V1 = {B2, B6, B7}**

View **V2 = {B1, B2, B6, B7}**

- The monotonicity property insists that if items are initially assigned to a set of buckets V_1 and then some new buckets are added to form V_2, then an item may move from an old bucket to a new bucket, but not from one old bucket to another old bucket.
- This reflects an intuition about consistency: when the set of usable buckets changes, items should only move if necessary to preserve an even distribution.

A Construction

View V1 = {B2, B6, B7}
View V2 = {B1, B3, B5}
View V3 = {B2, B6, B7}
View V4 = {B1, B2, B4}

v = 4

Object id, i = 12

$f(i, V1) = 1$
$f(i, V2) = 4$
$f(i, V3) = 1$
$f(i, V4) = 2$

$\sigma(12) = 3$

- We consider ranged hash families that have the following properties.
- Spread: Let $V_1, . . ., V_v$ be a set of views, altogether containing C distinct buckets and each individually containing at least C/t buckets.
- For a ranged hash function f and a particular item i, the spread $\sigma(i)$ is the quantity $|\{f_{V_j}(i)\}_{j=1}^{v}|$.

A Construction

View $V1 = \{B2, B6, B7\}$
View $V2 = \{B1, B3, B5\}$
View $V3 = \{B2, B6, B7\}$
View $V4 = \{B1, B2, B4\}$

$v = 4$

Object id, $i = 12$
$f(i, V1) = 1$
$f(i, V2) = 4$
$f(i, V3) = 1$
$f(i, V4) = 2$

$\sigma(12) = 3$

- Spread: Let $V_1, \ldots, V_v$ be a set of views, altogether containing C distinct buckets and each individually containing at least C/t buckets.
- For a ranged hash function f and a particular item i, the spread $\sigma(i)$ is the quantity $|\{f_{V_j}(i)\}_{j=1}^{v}|$.
- The spread of a hash function $\sigma(f) := \max_i \sigma(i)$
- The spread of a hash family is σ if, with high probability, the spread of a random hash function from the family is σ.

A Construction

- The idea behind spread is that there are v clients, each of which has a view that contains at least a constant fraction (1/t) of the buckets.
- Each client tries to assign an item i to a bucket using a consistent hash function.
- The property says that across the entire group, there are at most $\sigma(i)$ different opinions about which bucket should contain the item.
- Clearly, a good consistent hash function should have low spread over all items.

A Construction

View V1 = {B2, B6, B7}
View V2 = {B1, B3, B5}
View V3 = {B2, B6, B7}
View V4 = {B1, B2, B4}

Object Id	12	28	74	65
V1	2	7	6	2
V2	**1**	**3**	**3**	**5**
V3	7	6	2	6
V4	1	4	2	4

Load of B3 under V2 = 2.

- We consider ranged hash families that have the following properties.
- Load: Define a set of V views as before.
- For a ranged hash function f and a bucket b, the load $\lambda(b)$ is the quantity $|U_{v \text{ in } V} f_v^{-1}(b)|$.

A Construction

View V1 = {B2, B6, B7}
View V2 = {B1, B3, B5}
View V3 = {B2, B6, B7}
View V4 = {B1, B2, B4}

Object Id	12	28	74	65
V1	2	7	6	2
V2	**1**	**3**	**3**	**5**
V3	7	6	2	6
V4	1	4	2	4

Load of B3 under V2 = 2.

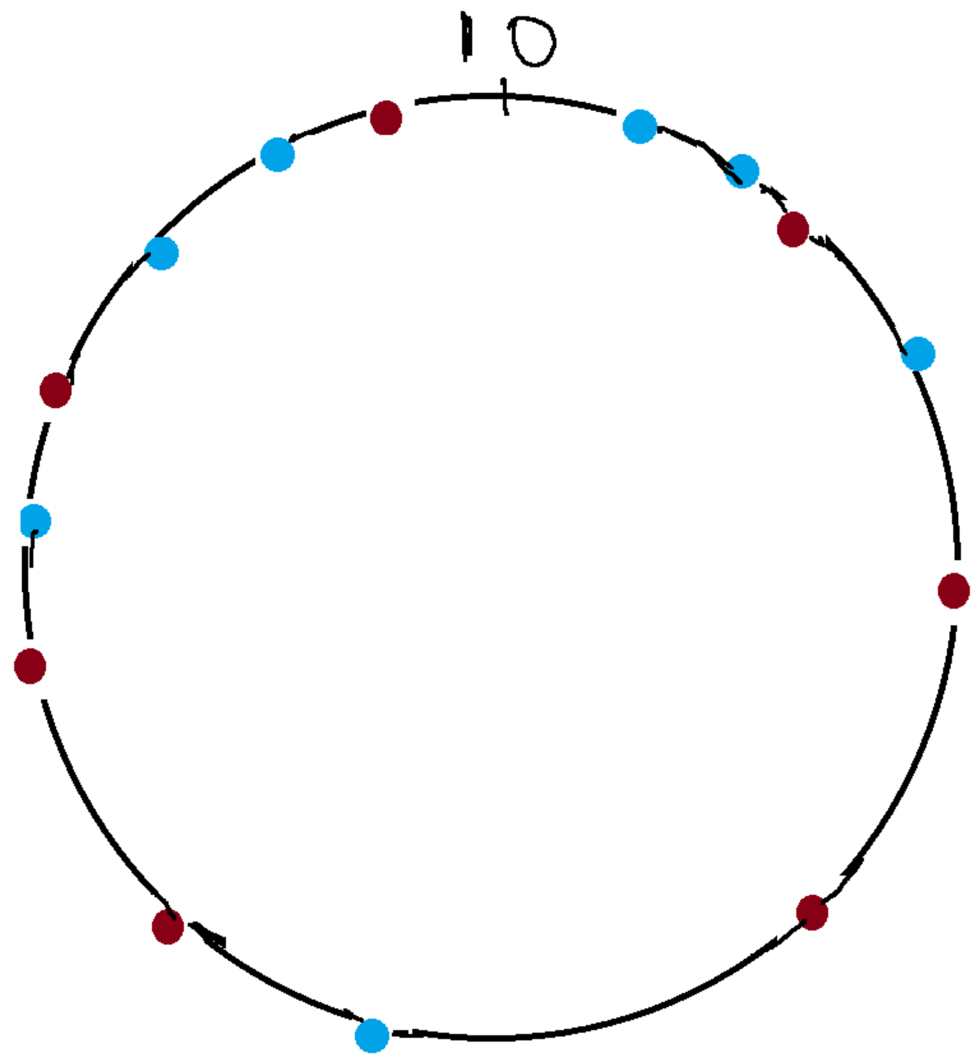
- Load: Define a set of V views as before.
- For a ranged hash function f and bucket b, the load $\lambda(b)$ is the quantity $|U_{v \text{ in } V} f_v^{-1}(b)|$.
- The load of a hash function $\lambda(f) = \max_b \lambda(b)$.
- The load of a hash family is λ if with high probability, a randomly chosen hash function has load λ, whp.

A Construction

- The load property is similar to spread.
- The same V buckets are back, but this time consider a particular bucket b instead of an item.
- The property says that there are at most $\lambda(b)$ distinct items that at least one client thinks belongs in the bucket.
- A good consistent hash function should also have low load.

A Construction

- Suppose that we have two random functions r_B and r_I.
- The function r_B maps buckets randomly to the unit interval, [0,1]
- The function r_I does the same for items.
- $f_v(i)$ is defined to be the bucket $b \in V$ that minimizes $|r_B(b) - r_I(i)|$.
- In other words, i is mapped to the bucket "closest" to i.



A Construction

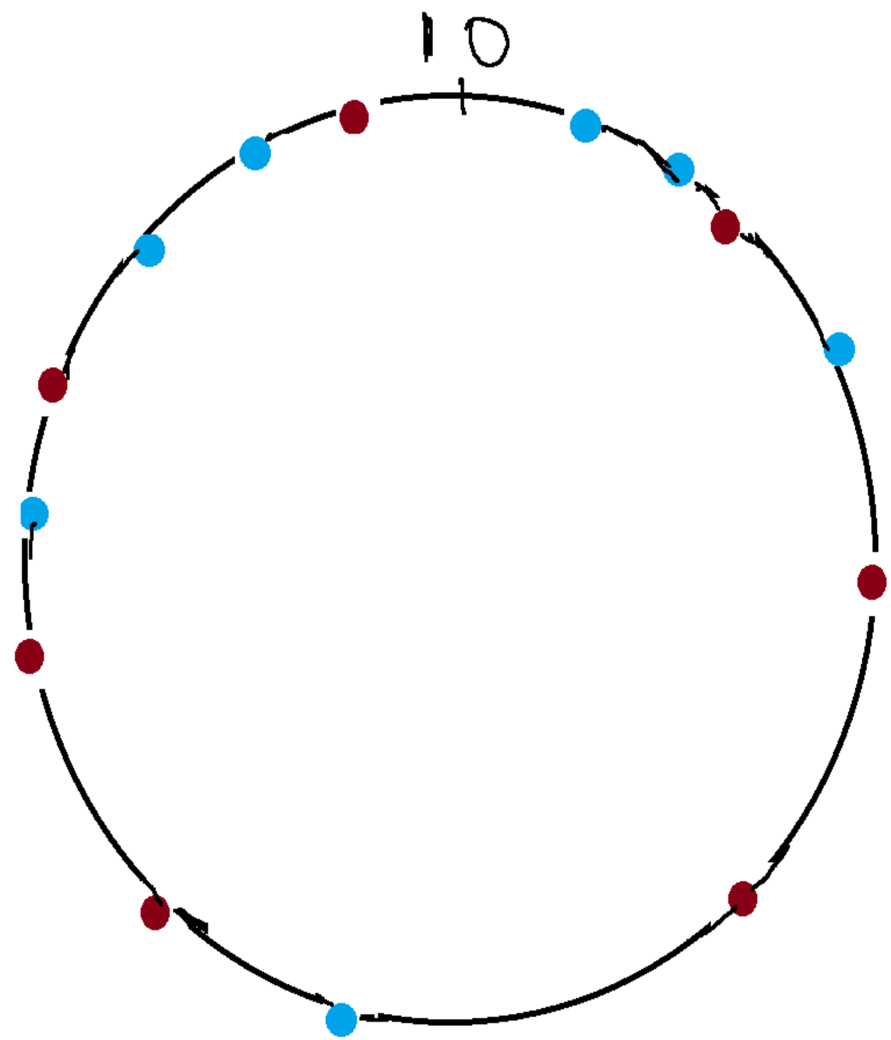
- For the purposes of analysis, we will have each bucket mapped to several points in [0, 1).
- If the number of buckets in the range is always less than C, we will use k.log(C) points for each bucket for some constant k.
- The easiest way to think of this arrangement is to consider replicating each bucket by klog(C) times.
- The function r_B maps each replicated bucket randomly.

A Construction

- The construction above is a ranged hash family that is monotone, has a spread of O(log C), and a load of O(log C), with high probability.
- C is the number of buckets.
- The proof largely relies on using Chernoff bounds and the independence of the functions r_B and r_I.
- Refer to the paper for details.

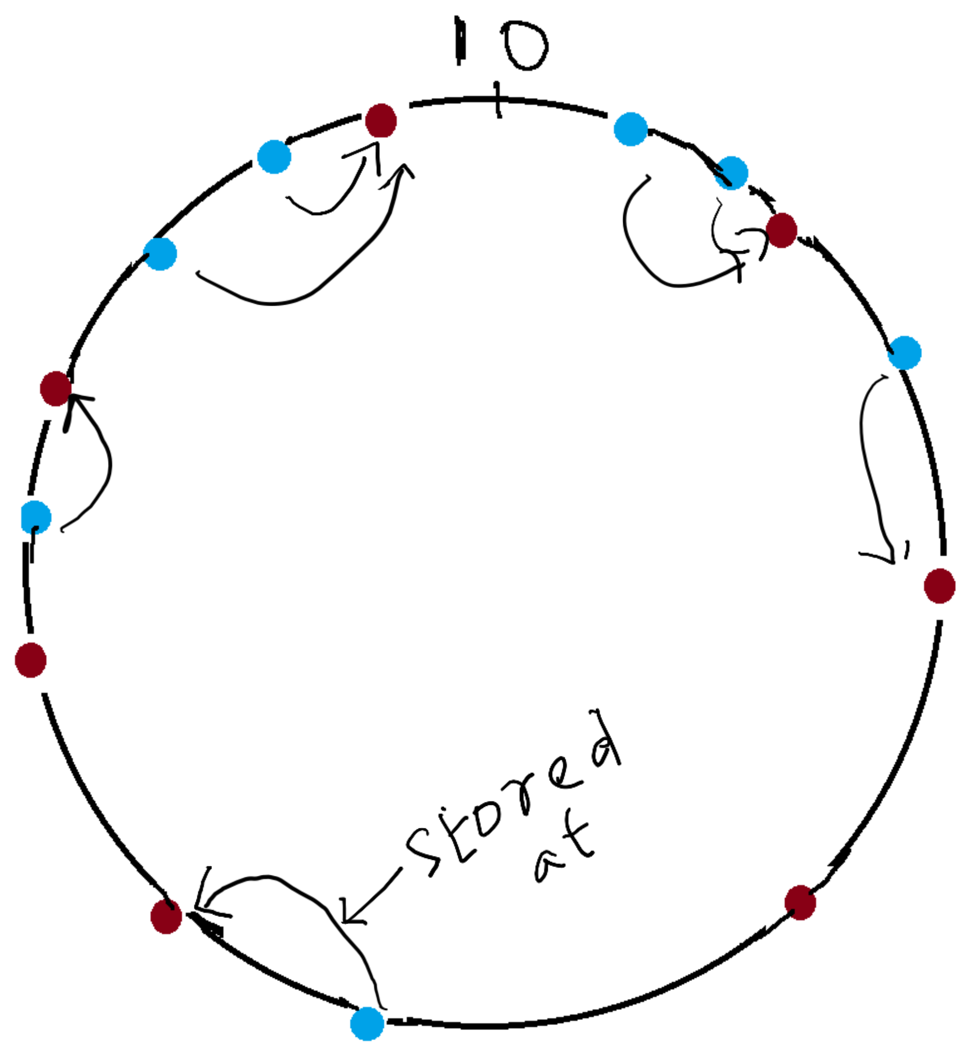
First Example – Chord

- Uses ideas from consistent hashing.
- Node address as well as object value is mapped to a logical identifier in a common space using a consistent hash function.
- When a node joins or leaves the network of n nodes, only 1/n items have to be moved.
- Two steps involved.
  - Map the object value to its key
  - Map the key to the node in the native address space using lookup



How to Place Data

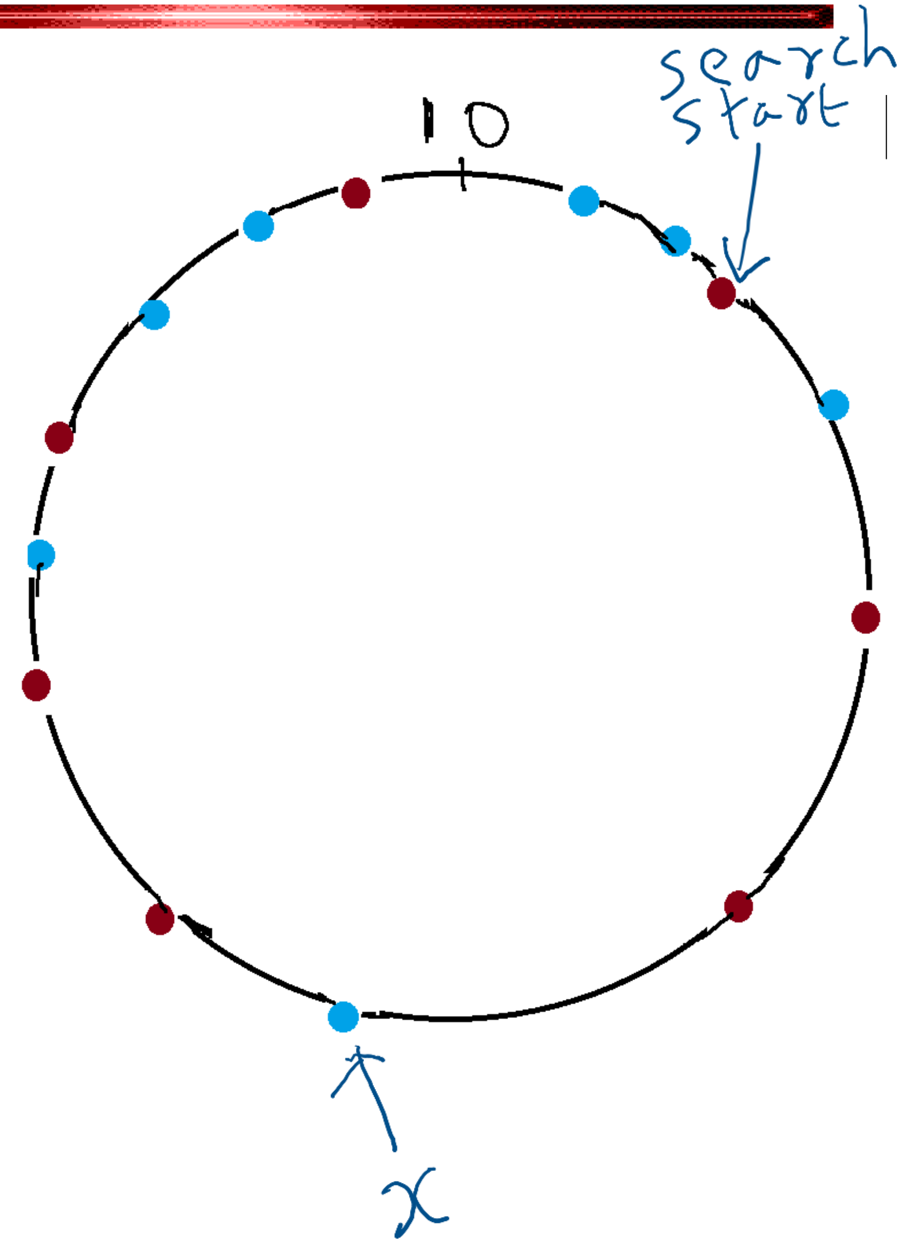
- Common address space is an m-bit identifier (2^m addresses), and this space is arranged on a logical ring mod(2^m).
- A key k gets assigned to the first node such that the node identifier equals or is greater than the key identifier k in the logical space address.



Consider $2^m = 1$. $k = 1.476293$, $k \bmod m = 0.476293$

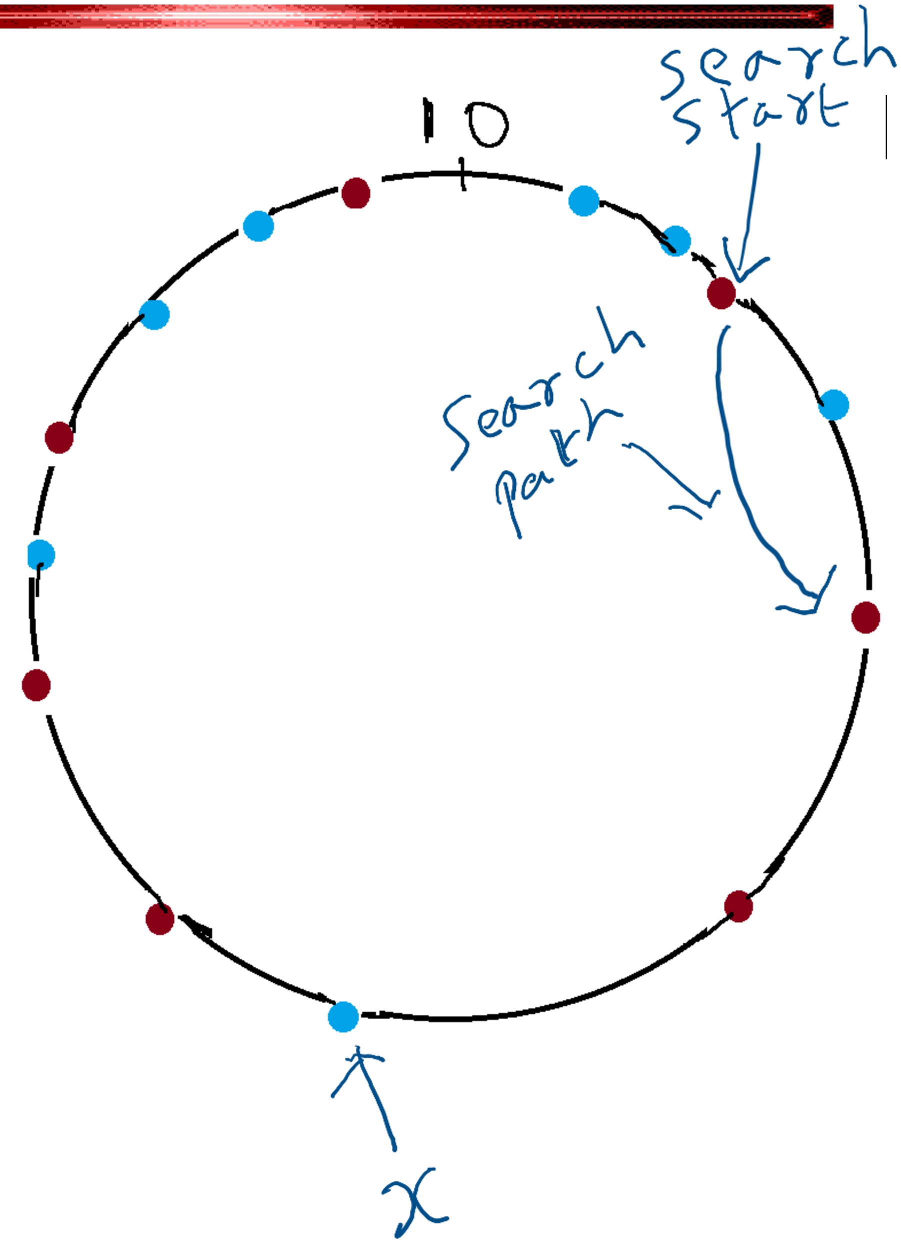
Chord – Lookup

- Each node tracks its successor on the ring.
- A query for key x is forwarded on the ring until it reaches the first node whose identifier y ≥ key x mod(2^m).
- The result is returned to the querying node on the reverse of the path that was followed by the query.
- This mechanism requires O(1) local space but O(n) hops.



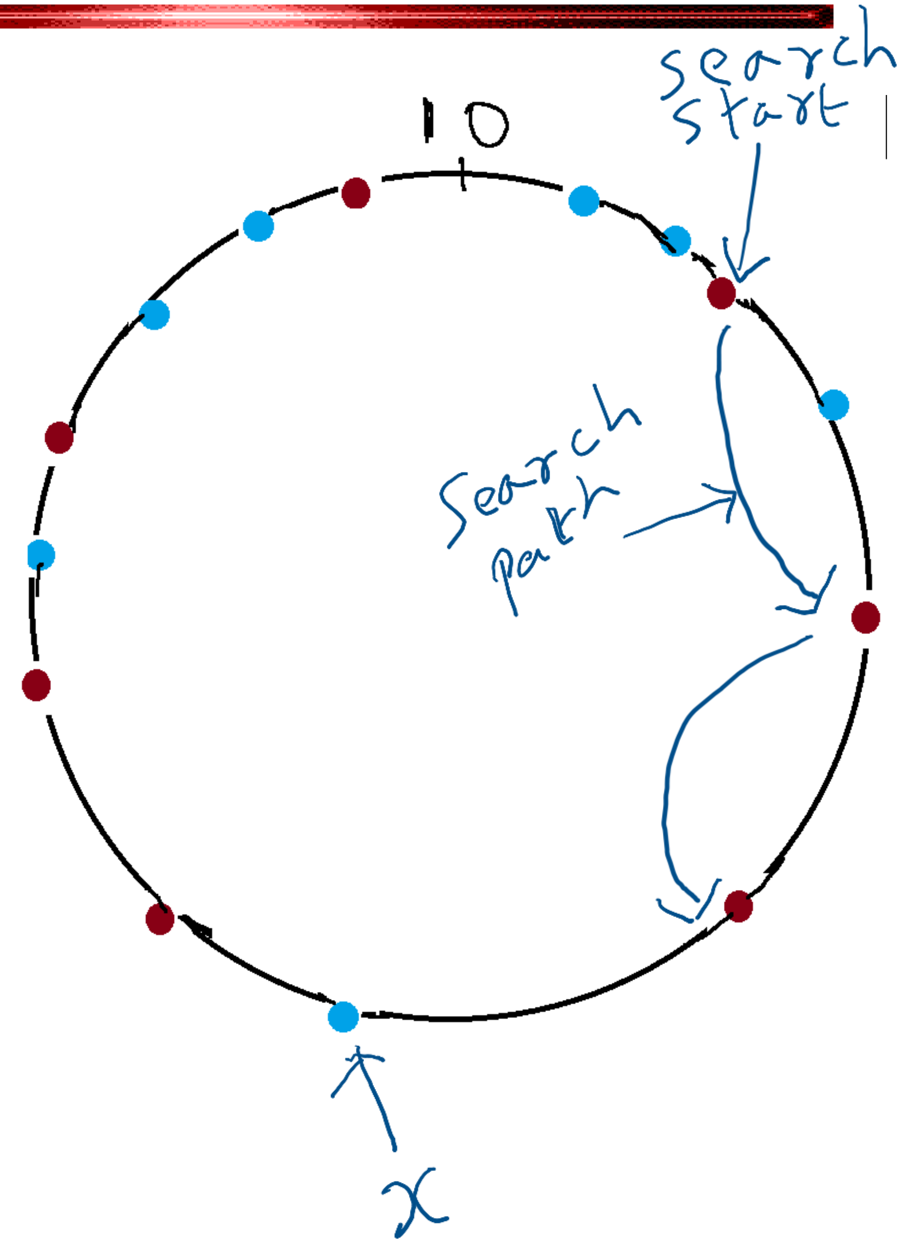
Chord – Lookup

- Each node tracks its successor on the ring.
- A query for key x is forwarded on the ring until it reaches the first node whose identifier y ≥ key x mod(2^m).
- The result is returned to the querying node on the reverse of the path that was followed by the query.
- This mechanism requires O(1) local space but O(n) hops.



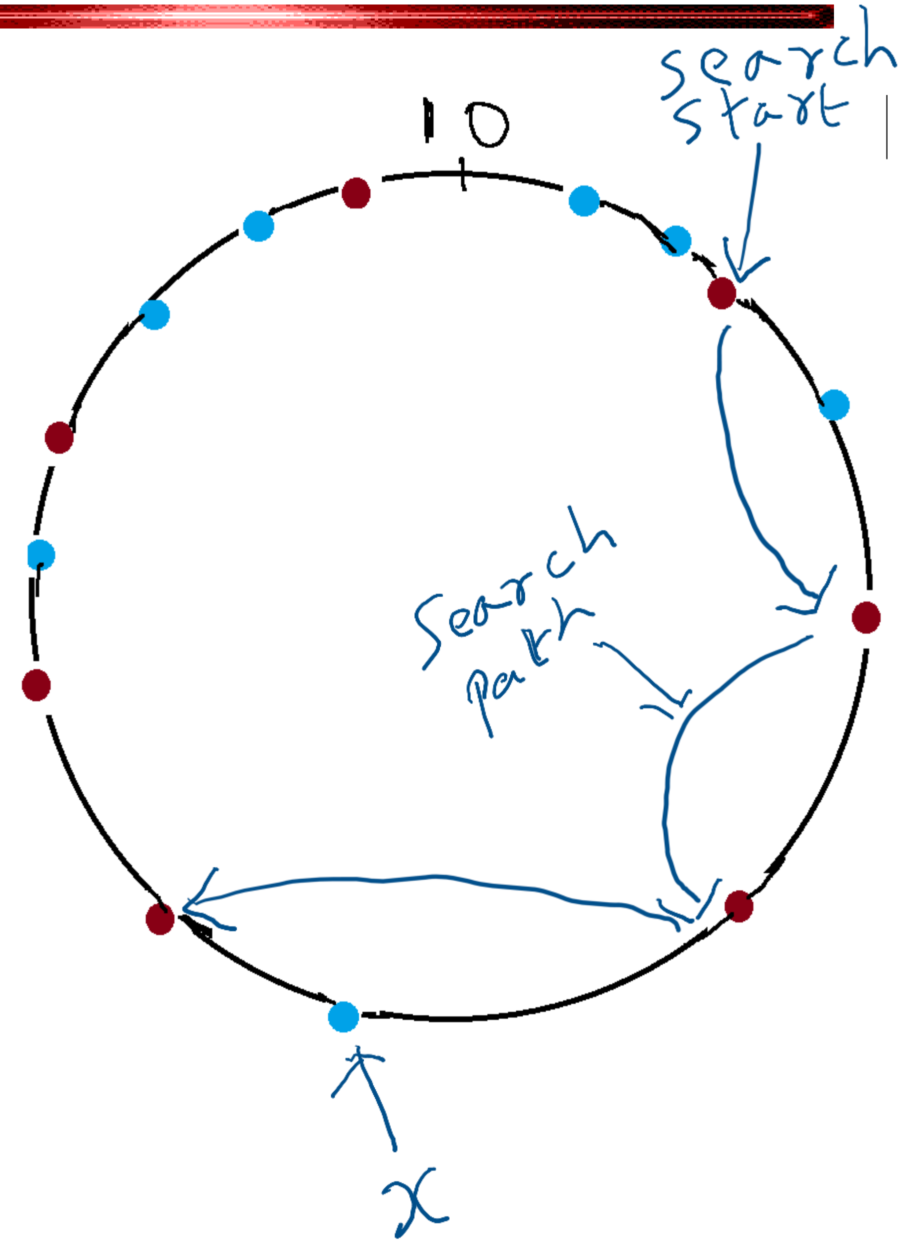
Chord – Lookup

- Each node tracks its successor on the ring.
- A query for key x is forwarded on the ring until it reaches the first node whose identifier y ≥ key x mod(2^m).
- The result is returned to the querying node on the reverse of the path that was followed by the query.
- This mechanism requires O(1) local space but O(n) hops.



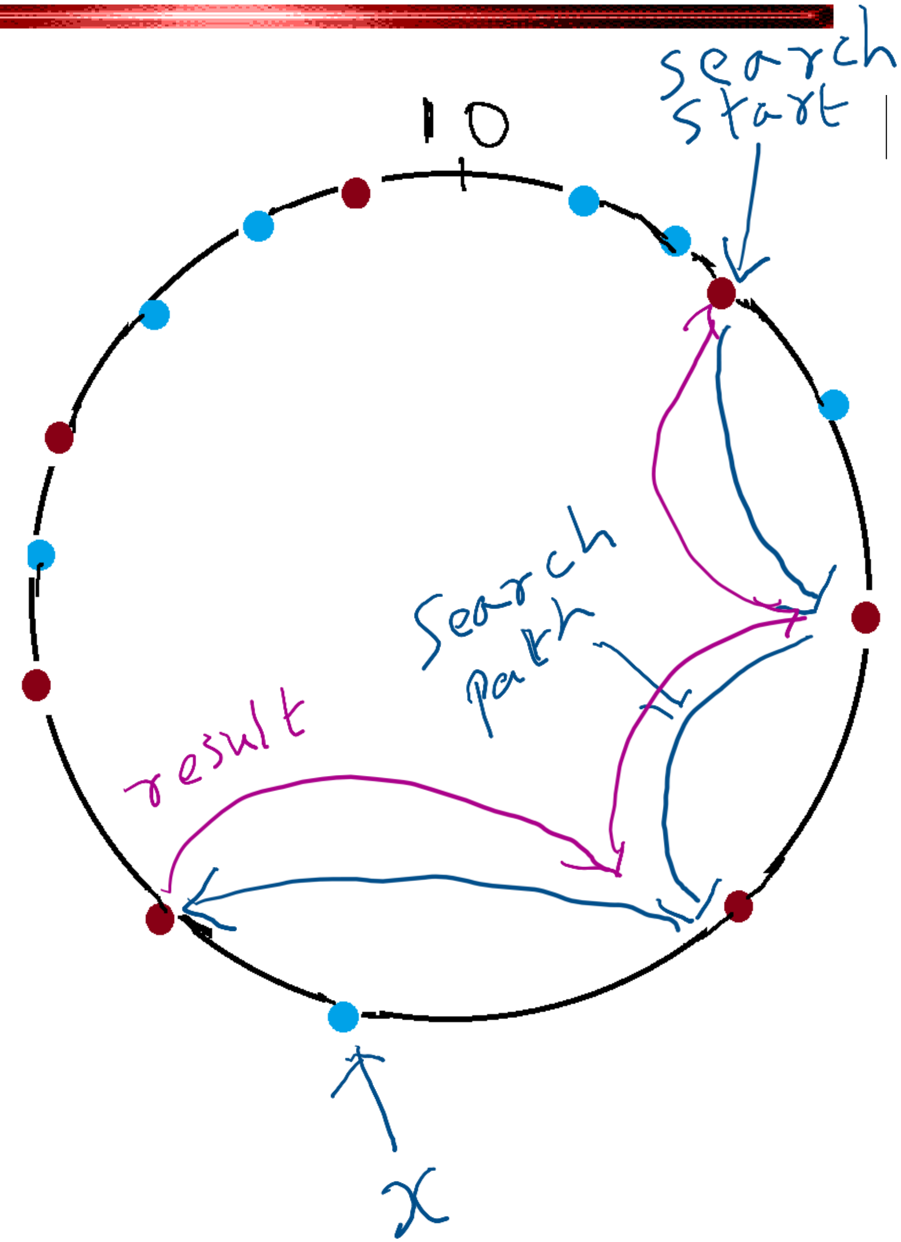
Chord – Lookup

- Each node tracks its successor on the ring.
- A query for key x is forwarded on the ring until it reaches the first node whose identifier y ≥ key x mod(2^m).
- The result is returned to the querying node on the reverse of the path that was followed by the query.
- This mechanism requires O(1) local space but O(n) hops.



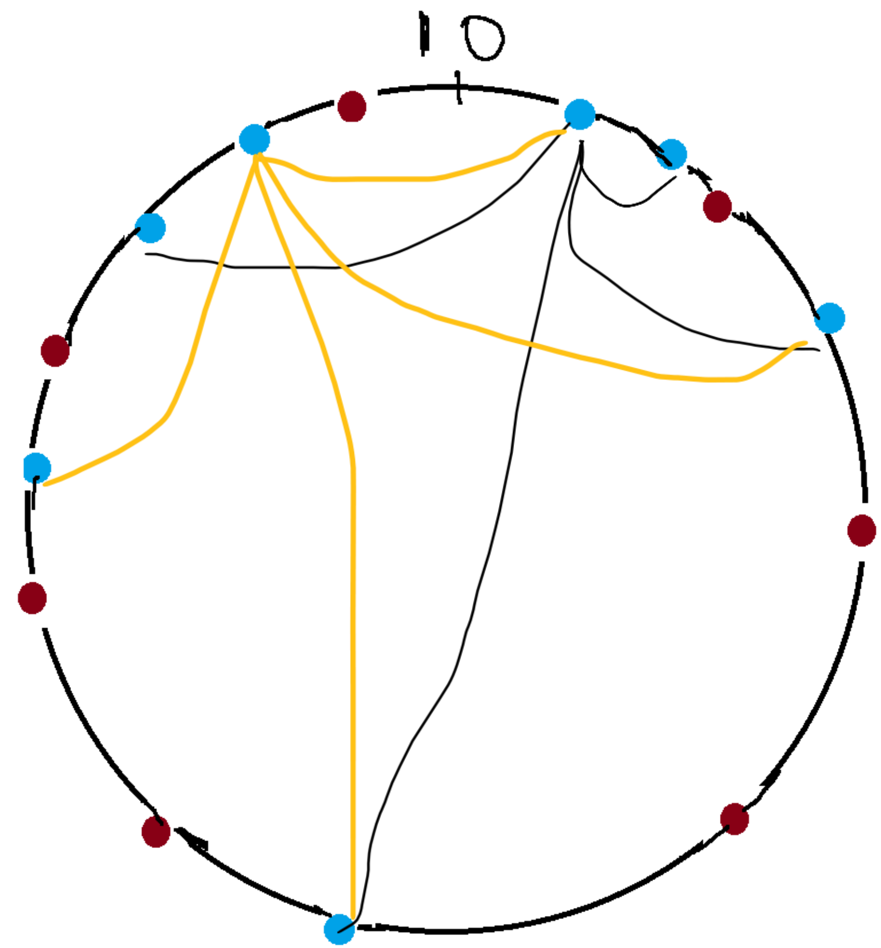
Chord – Lookup

- Each node tracks its successor on the ring.
- A query for key x is forwarded on the ring until it reaches the first node whose identifier y ≥ key x mod(2^m).
- The result is returned to the querying node on the reverse of the path that was followed by the query.
- This mechanism requires O(1) local space but O(n) hops.



Chord – Improved Lookup

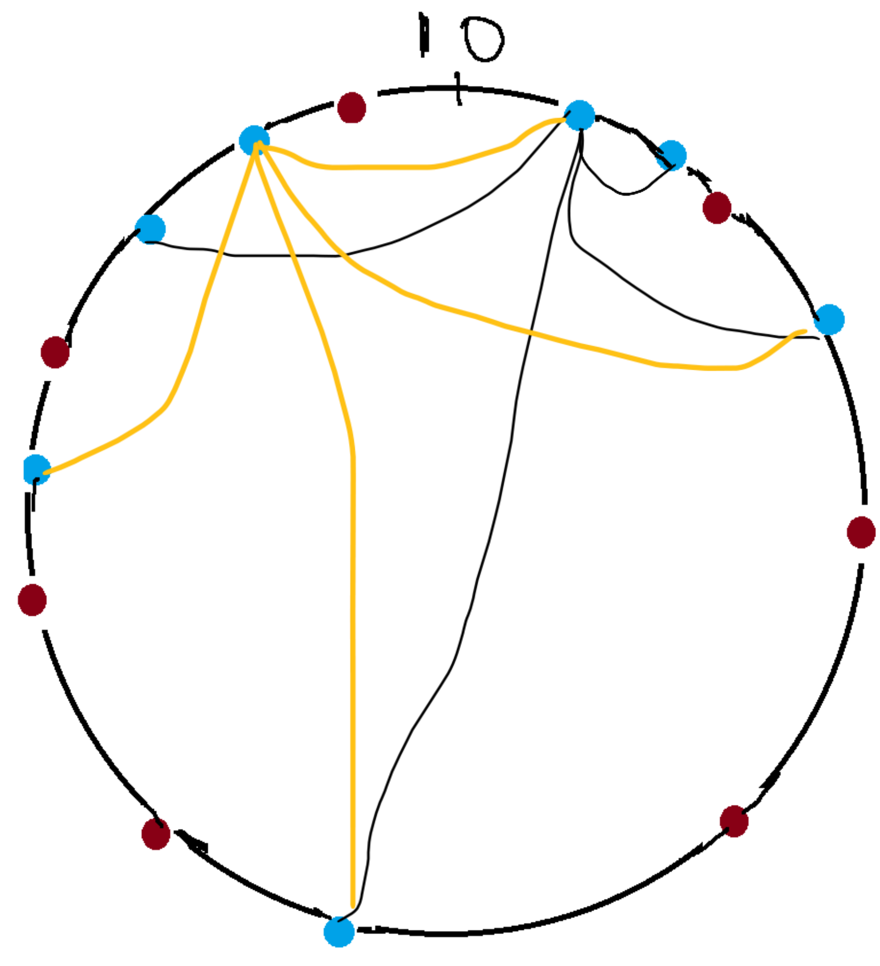
j	j^{th} Finger
1	(0.35 + 1/16) mod 1
2	(0.35 + 2/16) mod 1
3	(0.35 + 4/16) mod 1
4	(0.35 + 8/16) mod 1



- Consider a node at position i with i in [0, 1).
- Node i maintains a routing table, called the finger table.
- For $1 \le j \le m$, the j^{th} finger is a node at position $i + 2^{j-1-m}$.
- If there is no node at $i + 2^{j-1-m}$, use the successor of the node at $i + 2^{j-1-m}$.

Chord – Improved Lookup

- Can visualize the process of searching (routing) as a binary search.
- First, contact the peer via the finger table entry corresponding to x = 1.
- If this node is not the one that has the required file, then the next node searched is the finger table entry x = 2.
- Complexity: O(log n) message hops at the cost of O(log n) space in the local routing tables



Chord – Managing Churn

- Processing node join and leave
- When a new node joins, it has to:
  - Find its predecessor and successor
  - Own the items that will now be mapped to the new node.
  - Establish its finger table.
    - Can build its finger table by using the finger table of its successor.
- When a node leaves, it has to
  - Inform its predecessor and successor
  - Transfer items that it owns to its successor

Other Examples of Overlay Networks

- Other structured overlay network models include
  - CAN – Content Addressable Network
  - Tapestry
  - And many more

Gnutella – An Unstructured Overlay Network

- A joiner connects to some standard nodes from Gnutella directory
- Ping used to discover other hosts; allows new host to announce itself
- Pong in response to Ping; Pong contains IP, port #, max data size for download
- Query messages used for flooding search
- QueryHit are responses. If data is found, this message contains the IP, port#, file size, download rate, etc.
- Path used is reverse path of Query.
- Provisions for handling nodes staying behind a firewall or in a private network.

Gnutella – An Unstructured Overlay Network

- Semantic indexing possible : keyword, range, attribute-based queries
- Easily accommodate high churn
- Efficient when data is replicated in network
- Good if user satisfied with "best-effort" search
- Network is not so large as to cause high delays in search

Search in Gnutella

- Flooding: with checking, with TTL or hop count, expanding ring strategy
- Random Walk: k random walkers, with checking
- Relationships of interest
  - The success rate as a function of the number of message hops, or TTL.
  - The number of messages as a function of the number of message hops, or TTL.
  - The above metrics as the replication ratio and the replication strategy changes.
  - The node coverage, recall, and message efficiency, as a function of the number of hops, or TTL; and of various replication ratios and replication strategies.
- Overall, k-random walk outperforms flooding

BitTorrent

- An unstructured overlay network like Napster but is very popular and navigated the legal challenges carefully.
- The aim of bit torrent is to run a legal network and not circumvent local laws.
- Used to store and forward several types of files including video, OS images, open source packages, datasets, educational content, and so on.

BitTorrent

- For each file, a torrent descriptor file has to be created.
- This descriptor file is also knows as the torrent.
- Contains file metadata and an SHA-1 output of the file contents.
- Once a server hosts a file, it creates the torrent and shares the torrent with other servers.
- The torrent files are tracked by dedicated trackers by storing a mapping between the torrent and the locations where the file is available.

BitTorrent – Mechanism

- BitTorrent supports a swarm of hosts. Every host participates in the system.
- Big files are broken into multiple pieces, each piece is limited in size to 256 KB.
    - File is no longer an atomic unit.
    - This small sized pieces is supposed to improve network usage.
- The pieces are distributed across peers.
- Each peer hosts pieces of different files.

BitTorrent Mechanism

- Having the file at multiple hosts allows a client to download the pieces simultaneously from the hosts.
- The trackers coordinate the transfer of the file along with a mapping of the piece to the hosts.
- Users can use regular search mechanisms that includes Google searches to find torrent descriptor files.
- If a server has a new file, it hosts it. Subsequently, it distributes the torrent file to let client machines and trackers know about it.

BitTorrent – Mechanism

- Nodes in BitTorrent can use multiple strategies to promote the behavior of users in the network.
- The upload bandwidth from a host to a client can be a function of how the client participates in the network.

- Anonymity and privacy are not part of the BitTorrent network design and mechanism.
- The legal onus is on the site that indexes and hosts the torrents.

FreeNet – A Third Generation System

- One of the popular third generation distributed data system is Freenet.
- Recall that third generation systems have to ensure anonymity and privacy of both the sender and receiver of data/files.
- Freenet offers anonymity and deniability as well.
- Think of applications such as dark web, anonymous browsing and hosting.

Freenet Node

- Each node maintains an index and a routing table.
- Routing table entries store the address of some nodes in the network and the files (keys of files) they have.
- Queries are routed to neighbors using flooding with a Time-to-Live (TTL) parameter.
- Identifiers attached to queries so as to prevent cycles while forwarding via flooding.
- Once a node N finds a file, the node has to send the contents to the client.
- However, cannot reveal the identity of N or other intermediaries in the network.

File Transfer

- Anonymity by obfuscation is what is used in Freenet.
- The result of the query follows the same path as the original search request as in Gnutella.
- Each node on the way claims to be the owner of the file -- this hides the real owner.
- Every node on the way caches a copy of the file and creates an entry in its routing table with the key and id of the data source.

Freenet

- A positive side effect is that popular data will tend to get replicated at many nodes, and it is thus easy to locate and fetch it.
- However, on the flip side, routing tables will get bigger because new entries shall get created and added to routing tables all the time.

Summary

- Overlays have been popular due to their peer model
- Truly distributed in that sense
- Structured vs. unstructured
  - Structured based on strong theory
  - Can result in good bounds on routing
  - Consistent hashing is a big development, two decades ago.
- There are several attempts to unify the structure and other processes of structured overlay networks.